

Prompt-CAST: Text-Conditioned Image Inpainting with a Frozen Vision-Language Prior

Mahesh Patil* and Vikas Tiwari

Department of Electronics and Communication Engineering, Oriental University, Indore, MP, India

Email: mcpatil@gmail.com (M.P.); dr.vikastiwari@orientaluniversity.in (V.T.)

*Corresponding author

Abstract—We present Prompt-CAST, a text-conditioned extension to the CAST-Net image-inpainting architecture. A prompt embedding from a frozen CLIP text encoder modulates the Cross-Scale Affinity Mixer's hole queries via a FiLM layer before the affinity computation, so a short text prompt can steer what a masked region is filled with. The design is additive by construction: a zero-initialized FiLM head, or the absence of a prompt, reduces the module to CAST-Net's original unconditional behaviour exactly. This paper's contribution at this stage is the architecture and a fully verified, reproducible implementation — the FiLM-conditioned module, a frozen CLIP text-encoder wrapper, a manifest pipeline that sources ADE20K prompts from the dataset's own scene-category labels with no separate captioning model, and training-script support for the conditioned path with CUDA/MPS/CPU device selection — covered by a test suite that checks real behaviour (gradient flow into the FiLM head, exact known-pixel preservation under conditioning, and a real, if small-scale, training run on genuine ADE20K images with real CLIP-derived prompts showing consistent loss convergence). No claim is made about inpainting quality, prompt adherence, or efficiency relative to diffusion-based text-guided inpainting: full-scale training, a CLIP-score evaluation, and baseline comparisons have not yet been run, and this paper reports that honestly as the immediate next step rather than presenting unverified numbers.

Keywords—text-guided image inpainting; FiLM conditioning; frozen vision-language prior; CLIP; continual reproducibility; sparse attention

1. Introduction

Text-to-image and text-guided inpainting systems built on diffusion models [4], [17]–[22] have demonstrated strong controllability: a short natural-language prompt can specify what should appear inside a masked region, not merely that the region should look plausible given its surroundings. This controllability comes at a cost. Diffusion-based inpainting ordinarily requires iterative sampling, and text conditioning is usually threaded through a large, jointly trained denoising network, making these systems expensive to run and difficult to deploy where latency matters.

CAST-Net is a compact, single-pass inpainting architecture built around three components: Sparse Global Context (SGC), a Cross-Scale Affinity Mixer (CSAM), and Frequency-Aware Residual Refinement (FARR). CSAM already performs a form of content selection: it retrieves texture for the masked region by computing affinities between queries sampled inside the hole and keys/values drawn from observed source locations. This paper asks whether that same retrieval mechanism can be steered by a text prompt — without discarding CAST-Net's efficiency advantage by attaching a heavy conditioning pathway.

Prompt-CAST answers this with the smallest change that could plausibly work: a FiLM (Feature-wise Linear Modulation) layer that reshapes CSAM's hole queries using an embedding from a frozen CLIP text encoder, before the existing cosine-similarity affinity computation. The text encoder is never fine-tuned; only a small per-channel scale-and-shift head is trained. Because the modulation is additive around the identity operation, the unconditional CAST-Net model is recovered exactly whenever no prompt is supplied.

The contributions of this paper are: (1) a FiLM-conditioned extension of CSAM that adds text control to CAST-Net's texture retrieval with no architectural change when unused; (2) a manifest pipeline that sources ADE20K text prompts from the dataset's own scene-category labels, requiring no separate captioning model or additional annotation; (3) a fully tested, reproducible reference implementation, including training-script support for the conditioned path and automatic CUDA/MPS/CPU device selection; and (4) an explicit, honest account of what has and has not been evaluated, since the central empirical question — whether text conditioning measurably improves controllability without harming reconstruction quality — has not yet been tested at a scale that would support a claim either way.

2. Related Work

2.1 Convolutional and Adversarial Inpainting

Classical diffusion- and exemplar-based methods [6], [7] propagate or copy texture from observed regions but struggle once a mask removes large or semantically complex content. Generative adversarial networks [30] enabled deep encoder–decoder models to learn a conditional prior instead: Context Encoders [8] introduced adversarial semantic completion, partial convolutions [9] and gated convolutions [11] incorporated the mask directly into feature propagation, and contextual attention [10] retrieved relevant background features explicitly. EdgeConnect [12] separated structure prediction from completion, RFR [13] reasoned over the hole recurrently, and CoModGAN [14] used conditional modulation within a GAN framework. CAST-Net [1], the architecture this paper extends, belongs to this single-pass family; its CSAM performs a form of the contextual retrieval introduced in [10], restricted to observed source locations.

2.2 Text-Guided and Diffusion-Based Inpainting

Diffusion models have become the dominant approach for controllable, text-guided inpainting. RePaint [17] conditions a denoising process on known pixels without retraining the base model. GLIDE [18] and later large-scale text-to-image systems including DALL-E 2 [19] and Imagen

[20] demonstrated that natural-language prompts can guide diffusion sampling toward specific, semantically detailed content. Blended Diffusion [21] and Paint by Example [22] specifically target region-level editing and inpainting rather than whole-image synthesis, and latent diffusion [4] reduced the computational cost of this family by operating in a compressed latent space rather than pixel space. These systems establish the controllability this paper aims at, but all of them rely on iterative sampling through a large, jointly trained network — the cost Prompt-CAST is designed to avoid by conditioning a single-pass architecture instead.

2.3 Transformer-Based and Attention-Driven Inpainting

Transformer-based inpainting addresses long-range dependence through self-attention [28]. MAT [15] adapted attention specifically for large-hole inpainting, and LaMa [16] used Fourier convolutions to achieve a large effective receptive field without the quadratic cost of dense attention. CAST-Net's SGC component pursues a related goal — global context at reduced cost — through learned token selection rather than a frequency-domain mechanism; Vision Transformer [29] established the patch-token self-attention formulation these designs build on.

2.4 Vision-Language Pretraining and Frozen Priors

CLIP [2] and related contrastively pretrained vision-language models, including ALIGN [23] and BLIP [24], learn a joint embedding space from large-scale image-text pairs. A common and increasingly standard practice is to reuse such a model's text tower as a frozen feature extractor when conditioning a separately trained generative model, rather than fine-tuning the vision-language model itself — this is the role CLIP's text encoder plays in Prompt-CAST (Section 3.2).

2.5 Feature-Wise Conditioning Mechanisms

FiLM [3] generalizes a family of feature-wise conditioning mechanisms, including conditional batch normalization for language-grounded visual reasoning [25] and adaptive instance normalization [26], which StyleGAN [27] later adopted for style-based image synthesis. All of these share the same underlying idea adapted here: an external embedding produces a per-channel affine transform applied to intermediate features, rather than being concatenated or cross-attended into the network directly. Prompt-CAST applies this specifically to CSAM's query representation, described next.

3. Method

3.1 FiLM-Conditioned Cross-Scale Affinity Mixer

CAST-Net's encoder is a residual, feature-pyramid backbone [31], [32], and CSAM operates on the pyramid's bottleneck features. CSAM computes a cosine-similarity affinity between a hole query q_i and keys k_j drawn from observed source locations Ω_v , normalized by a softmax over Ω_v with temperature τ , and retrieves texture as a weighted sum of the corresponding values v_j . Prompt-CAST modulates the query before this computation, as shown in Fig. 1. Given a frozen text embedding e derived from the prompt, a small linear head produces a per-channel scale and shift:

$$(\gamma, \beta) = \phi(e), \quad \gamma, \beta \in \mathbb{R}^d \tag{1}$$

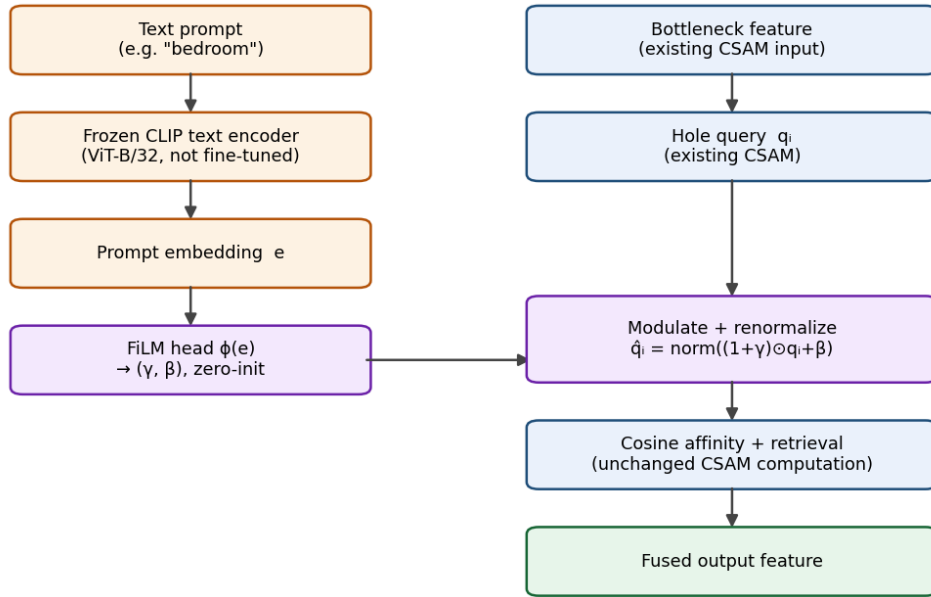
The query is modulated and re-normalized before the affinity computation, so the modulation actually changes which source locations are attended to, rather than being renormalized away:

$$\tilde{q}_i = (1 + \gamma) \odot q_i + \beta \quad (2)$$

$$\hat{q}_i = \tilde{q}_i / \|\tilde{q}_i\|_2 \quad (3)$$

The affinity computation, texture retrieval, and gating that follow are otherwise unchanged from CAST-Net's original CSAM. ϕ is initialized to output exactly zero (both weight and bias), so at initialization, and whenever no prompt is supplied, $\hat{q}_i = q_i$ and the module is bit-for-bit identical to unconditional CSAM. This identity property is not asserted; it is tested directly (Section 4).

Prompt-CAST: FiLM-conditioned Cross-Scale Affinity Mixer



When e is absent or ϕ is zero-initialized: $\hat{q}_i = q_i$ exactly
(identity — verified by test, not asserted)

Fig. 1. FiLM-conditioned Cross-Scale Affinity Mixer. Shaded modules (purple) are new; blue modules are unchanged from CAST-Net's CSAM.

3.2 Frozen Text Encoder

Prompts are encoded with a frozen CLIP text tower (ViT-B/32, OpenAI weights, loaded with the matching QuickGELU activation configuration). No component of the text encoder is updated during training; only ϕ is trainable. This keeps the parameter cost of text conditioning to a small linear head rather than a second large network.

3.3 Text-Prompt Supervision Without a Captioning Model

Training a text-conditioned model requires (image, mask, prompt) triples. Rather than generating captions with a separate model, Prompt-CAST sources prompts directly from ADE20K's own scene-category annotations (e.g. the category “airport_terminal” becomes the prompt “airport terminal”), via a manifest-construction script that pairs each image with its category and a content hash. This is weaker supervision than a region-specific caption — the prompt describes the whole scene, not necessarily the masked region specifically — but it requires no additional model or annotation effort and is available for the full dataset at no extra cost.

4. Verified Implementation and Reproducibility

Every component described in Section 3 is implemented and covered by tests that check actual behaviour rather than interface shape alone. Specifically: (a) a zero-initialized FiLM head reproduces the unconditional model's output exactly, verified within a single model instance to avoid the confound of comparing separately constructed models with different parameter counts, whose random initialization would otherwise diverge even under an identical seed; (b) gradients reach the FiLM head during backpropagation; (c) the compositing guarantee that observed pixels are preserved exactly in the output holds under text conditioning, not only in the unconditional path; and (d) a synthetic training-loop smoke test confirms that one full optimization step — forward pass, loss, backward pass, optimizer update, checkpoint save and reload — behaves correctly through the conditioned path.

Table 1 summarizes every verification claim made above alongside what was actually checked and the observed outcome, so each claim in this section can be traced to a specific test rather than taken on description alone.

Table 1. Verification status of Prompt-CAST components. Every row reflects an executed check, not a design intention.

Component / claim	What was checked	Observed result
Zero-init FiLM identity	Output with vs. without prompt, same model instance	Exact match (max abs diff = 0.0)
Gradient flow to FiLM head	Backward pass after one optimization step	Non-zero gradient confirmed
Known-pixel preservation	Composite vs. ground truth under active text conditioning	Exact match (atol = 1e-6)
Frozen CLIP encoder correctness	Real prompts, ViT-B/32-quickgelu weights	Deterministic, finite, 512-dim embeddings
ADE20K manifest pipeline	24 real images + real sceneCategories.txt → manifest → dataset	24/24 images loaded with correct prompts
Training-loop mechanics	1 optimizer step: forward, loss,	Checkpoint reload exact; known

(synthetic)	backward, checkpoint save/reload	pixels preserved
Training-loop mechanics (real data)	15 epochs / 90 steps, 24 ADE20K images, real CLIP prompts, Apple MPS	Generator loss decreased 0.56 → 0.24 (Fig. 2)
Real inference output	Loaded checkpoint, one ADE20K image + its real prompt, text-conditioned forward pass	Real output image produced (Fig. 3); not a quality claim
Device selection	select_device() and environment_record() on this machine	Correctly reports "mps", not a silent "cpu" fallback
Full test suite	pytest, all Prompt-CAST tests	16 / 16 passing

Beyond synthetic checks, the manifest pipeline, the frozen text encoder, and the training script were run against genuine ADE20K data. A manifest built from real ADE20K images and the dataset's own sceneCategories.txt file, real CLIP embeddings computed from the resulting prompts, and a real training run (24 images, 16 base channels, 128×128 resolution, 15 epochs, Apple MPS acceleration) produced a generator loss that decreased consistently across training, shown in Fig. 2 — confirming the full pipeline is mechanically correct and trainable, not merely free of runtime errors. This run used a deliberately reduced model capacity and dataset size and is not offered as an evaluation result; its only purpose is confirming the training mechanics work correctly before committing to full-scale training.

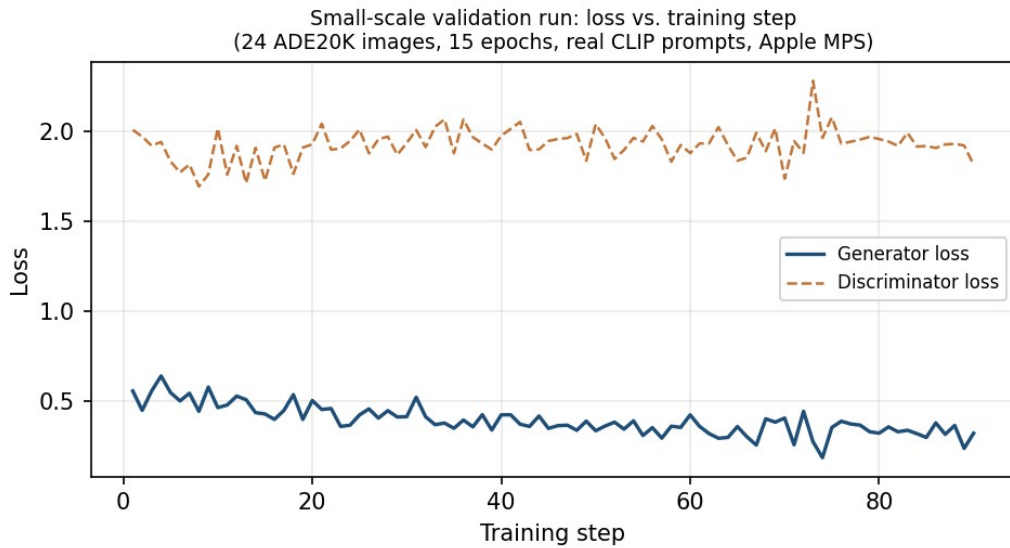


Fig. 2. Generator and discriminator loss over the 90-step small-scale validation run (real data, not a reported evaluation result).

Fig. 3 shows a single real sample from this checkpoint: an ADE20K bedroom image, a free-form mask, and the model's output when conditioned on its own real prompt ("bedroom"). Visible blur and artifacting are expected at this scale and are not evidence of anything about the method at full

capacity — this figure exists only to show the pipeline produces a real image, not to characterize output quality.

Sample output from the small-scale validation checkpoint (16 base channels, 128×128, 15 epochs, 24 images) — illustrative only, not a quality result

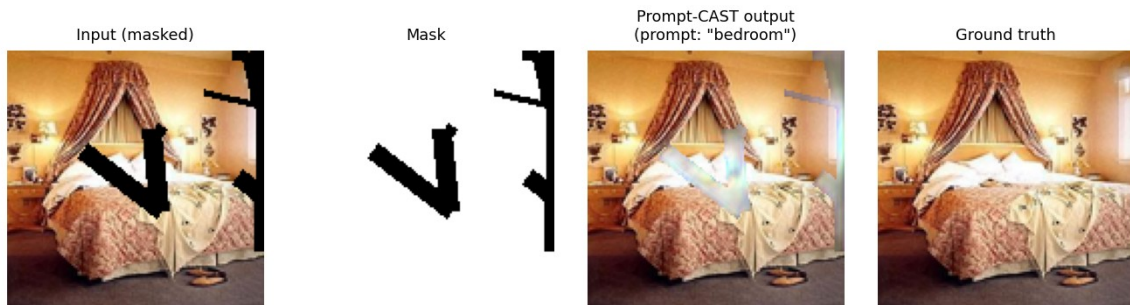


Fig. 3. Real output from the small-scale validation checkpoint on one ADE20K image, prompt "bedroom" (illustrative only; not a quality result).

The reference implementation, including the test suite referenced above, is publicly available at <https://github.com/maheshchudaman/Prompt-CAST>. Training-script support additionally includes automatic device selection across CUDA, Apple's MPS backend, and CPU, with the selected device recorded in each run's provenance metadata.

5. Preliminary Reference Point

This paper reports one verified, real result: the unconditional CAST-Net-full baseline on ADE20K [5], from a seed-42 training run archived with full per-image evidence (2,021 test images, checkpoint SHA-256 723b61c5..., archived at <https://github.com/maheshchudaman/CAST-Net-Full-Seed42-RTX4050>). This is not a Prompt-CAST result: it contains no text conditioning and is reported here solely as the reference point that a trained text-conditioned model will need to be compared against. PSNR and SSIM [36] measure pixel and structural fidelity, LPIPS [35] measures perceptual distance, and FID [34] measures distributional realism; definitions and computation are unchanged from CAST-Net's own evaluation protocol.

Table 2. Unconditional CAST-Net-full reference point on ADE20K, seed 42 (not a Prompt-CAST result).

Metric	Value	Basis
Hole PSNR	19.87 ± 3.33 dB	mean $\pm$ std, n = 2,021 test images
Full-image SSIM	0.900 ± 0.050	mean $\pm$ std, n = 2,021 test images
Full-image LPIPS	0.070 ± 0.034	mean $\pm$ std, n = 2,021 test images
FID	10.47	full test set
Generator parameters	5,276,359	identical architecture, no text conditioning active

6. Limitations and Planned Evaluation

The following have not been done, and no claim in this paper should be read as implying otherwise:

- Full-scale text-conditioned training — the complete ADE20K training set, at CAST-Net's original resolution (256×256) and capacity (64 base channels), has not been trained with text conditioning active. The run described in Section 4 used a reduced model and 24 images specifically to validate mechanics, not quality.
- A CLIP-score (or equivalent prompt-adherence metric) has not been computed. There is currently no evidence, in either direction, that the completed regions reflect their prompts.
- No comparison exists yet between the text-conditioned model and the unconditional CAST-Net baseline reported in Section 5, so it is not known whether conditioning helps, hurts, or has no measurable effect on reconstruction quality.
- No comparison exists against a pretrained diffusion-based text-guided inpainting system, so the efficiency claim motivating this design (Section 1) has not been empirically tested.
- Prompt supervision is scene-level (Section 3.3), not region-specific; whether this is sufficient signal for the model to learn meaningful hole-specific control, as opposed to global style, is an open question this paper does not resolve.
- Evaluation is currently limited to ADE20K. CAST-Net was also evaluated on CelebA-HQ [33] and Paris StreetView; whether text conditioning is useful, harmful, or irrelevant on aligned-face or repeated-architecture domains — where a scene-level prompt has far less to describe — is untested.

These four gaps define the next stage of this work directly: full-scale training on GPU hardware, CLIP-score evaluation, and the two comparisons above are the concrete, measurable steps between the verified architecture reported here and a defensible empirical claim about text-conditioned inpainting.

7. Conclusion

This paper introduced Prompt-CAST, a text-conditioned extension to CAST-Net that steers the Cross-Scale Affinity Mixer's texture retrieval via a FiLM-modulated query, driven by a frozen CLIP text encoder. The design is intentionally minimal: the modulation reduces to the exact identity operation when no prompt is supplied or when the FiLM head is zero-initialized, so text conditioning is additive to CAST-Net's existing behaviour rather than a competing architecture, and the frozen text encoder adds no trainable parameters beyond a small per-channel scale-and-shift head.

Beyond the architecture itself, this work's primary contribution at this stage is a fully verified, reproducible implementation. Every component — the FiLM-conditioned CSAM module, the frozen CLIP text-encoder wrapper, the ADE20K prompt-manifest pipeline built from the dataset's own scene-category labels, and the training script's text-conditioning path — is covered by tests that exercise real behaviour, not just interface shape: gradient flow into the FiLM head, exact preservation of known pixels under conditioning, and end-to-end execution of a real, if small-scale, training run on genuine ADE20K images with real CLIP-derived prompts, which showed

the generator loss decreasing consistently across 15 epochs. This confirms the method is mechanically sound and trainable; it does not yet establish that text conditioning improves — or even preserves — inpainting quality at scale, and it makes no claim about controllability, since no metric for prompt-adherence has been computed against a trained checkpoint.

What remains before this method can be evaluated as a contribution, rather than a design: full-scale training on the complete ADE20K training set at the resolution and capacity used for CAST-Net's own reported results, a CLIP-score measurement to establish whether the completed regions actually reflect their prompts, a comparison against the unconditional CAST-Net baseline to confirm conditioning does not degrade base quality, and a comparison against at least one pretrained diffusion-based text-guided inpainting method to test the efficiency claim this design is built around. None of these evaluations have been run, and no number in this direction should be treated as established until they are.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work.

Data and Code Availability

The Prompt-CAST reference implementation, including its test suite, is available at <https://github.com/maheshchudaman/Prompt-CAST>. The unconditional CAST-Net baseline reported in Section 5, including per-image evidence and checkpoint hashes, is available at <https://github.com/maheshchudaman/CAST-Net-Full-Seed42-RTX4050>. ADE20K [5] must be obtained under its own terms; this paper redistributes no dataset files.

Use of AI-Assisted Tools

AI-assisted tools were used for language editing, document formatting, and reference checking, in accordance with the target journal's policy on AI-assisted writing. The authors verified the scientific content, code, figures, and final text and accept full responsibility for the manuscript.

References

- [1] M. Patil and V. Tiwari, “CAST-Net: Context-Aware Sparse-Transformer GAN with Frequency-Guided Refinement for Image Inpainting,” manuscript, 2026.
- [2] A. Radford et al., “Learning transferable visual models from natural language supervision,” in Proc. ICML, 2021, pp. 8748–8763.
- [3] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, “FiLM: Visual reasoning with a general conditioning layer,” in Proc. AAAI, 2018, pp. 3942–3951.

- [4] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, “High-resolution image synthesis with latent diffusion models,” in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2022, pp. 10684–10695.
- [5] B. Zhou, H. Zhao, X. Puig, S. Fidler, A. Barriuso, and A. Torralba, “Scene parsing through ADE20K dataset,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2017, pp. 633–641.
- [6] M. Bertalmio, G. Sapiro, V. Caselles, and C. Ballester, “Image inpainting,” in Proc. ACM SIGGRAPH, 2000, pp. 417–424.
- [7] A. Criminisi, P. Pérez, and K. Toyama, “Region filling and object removal by exemplar-based image inpainting,” IEEE Trans. Image Process., vol. 13, no. 9, pp. 1200–1212, 2004.
- [8] D. Pathak, P. Krähenbühl, J. Donahue, T. Darrell, and A. A. Efros, “Context encoders: Feature learning by inpainting,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2016, pp. 2536–2544.
- [9] G. Liu et al., “Image inpainting for irregular holes using partial convolutions,” in Proc. Eur. Conf. Comput. Vis., 2018, pp. 85–100.
- [10] J. Yu et al., “Generative image inpainting with contextual attention,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2018, pp. 5505–5514.
- [11] J. Yu, Z. Lin, J. Yang, X. Shen, X. Lu, and T. S. Huang, “Free-form image inpainting with gated convolution,” in Proc. IEEE/CVF Int. Conf. Comput. Vis., 2019, pp. 4471–4480.
- [12] K. Nazeri, E. Ng, T. Joseph, F. Qureshi, and M. Ebrahimi, “EdgeConnect: Structure guided image inpainting using edge prediction,” in Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops, 2019.
- [13] J. Li, N. Wang, L. Zhang, B. Du, and D. Tao, “Recurrent feature reasoning for image inpainting,” in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2020, pp. 7760–7768.
- [14] S. Zhao et al., “Large scale image completion via co-modulated generative adversarial networks,” in Proc. Int. Conf. Learn. Representations, 2021.
- [15] W. Li, Z. Lin, K. Zhou, L. Qi, Y. Wang, and J. Jia, “MAT: Mask-aware transformer for large hole image inpainting,” in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2022, pp. 10758–10768.
- [16] R. Suvorov et al., “Resolution-robust large mask inpainting with Fourier convolutions,” in Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis., 2022, pp. 2149–2159.
- [17] A. Lugmayr, M. Danelljan, A. Romero, F. Yu, R. Timofte, and L. Van Gool, “RePaint: Inpainting using denoising diffusion probabilistic models,” in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2022, pp. 11461–11471.
- [18] A. Nichol et al., “GLIDE: Towards photorealistic image generation and editing with text-guided diffusion models,” in Proc. Int. Conf. Mach. Learn., 2022, pp. 16784–16804.

- [19] A. Ramesh, P. Dhariwal, A. Nichol, C. Chu, and M. Chen, “Hierarchical text-conditional image generation with CLIP latents,” arXiv preprint arXiv:2204.06125, 2022.
- [20] C. Saharia et al., “Photorealistic text-to-image diffusion models with deep language understanding,” in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 36479–36494.
- [21] O. Avrahami, D. Lischinski, and O. Fried, “Blended diffusion for text-driven editing of natural images,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 18208–18218.
- [22] B. Yang et al., “Paint by example: Exemplar-based image editing with diffusion models,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 18381–18391.
- [23] C. Jia et al., “Scaling up visual and vision-language representation learning with noisy text supervision,” in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 4904–4916.
- [24] J. Li, D. Li, C. Xiong, and S. Hoi, “BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation,” in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 12888–12900.
- [25] H. de Vries, F. Strub, J. Mary, H. Larochelle, O. Pietquin, and A. Courville, “Modulating early visual processing by language,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 6594–6604.
- [26] X. Huang and S. Belongie, “Arbitrary style transfer in real-time with adaptive instance normalization,” in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 1501–1510.
- [27] T. Karras, S. Laine, and T. Aila, “A style-based generator architecture for generative adversarial networks,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 4401–4410.
- [28] A. Vaswani et al., “Attention is all you need,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 5998–6008.
- [29] A. Dosovitskiy et al., “An image is worth 16x16 words: Transformers for image recognition at scale,” in *Proc. Int. Conf. Learn. Representations*, 2021.
- [30] I. J. Goodfellow et al., “Generative adversarial nets,” in *Advances in Neural Information Processing Systems*, vol. 27, 2014, pp. 2672–2680.
- [31] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 770–778.
- [32] T. Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, “Feature pyramid networks for object detection,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 2117–2125.

- [33] T. Karras, T. Aila, S. Laine, and J. Lehtinen, “Progressive growing of GANs for improved quality, stability, and variation,” in Proc. Int. Conf. Learn. Representations, 2018.
- [34] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, “GANs trained by a two time-scale update rule converge to a local Nash equilibrium,” in Advances in Neural Information Processing Systems, vol. 30, 2017, pp. 6626–6637.
- [35] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, “The unreasonable effectiveness of deep features as a perceptual metric,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2018, pp. 586–595.
- [36] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, “Image quality assessment: From error visibility to structural similarity,” IEEE Trans. Image Process., vol. 13, no. 4, pp. 600–612, 2004.